

Proposition Logic

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Proposition were given statements that results either 'True' or 'False'. Consider variables A, B, C,... which can have two values either 'True' or 'False'

K = 'Kritin is going to school' $\begin{matrix} \nearrow \text{True} \\ \searrow \text{False} \end{matrix}$

There are 3 logical operators:-

1. not $\neg, \bar{}, \neg$ $\neg A$
2. and $\wedge$
3. or $\vee$

Negation or Complement $\% \quad \text{not}(\text{not}(A)) = A, \quad \text{not}(1) = 0, \quad \text{not}(0) = 1$

$\text{not}(A) = A', \quad \text{not}(A') = A$

Associative $\% \quad (A \text{ and } B) \text{ and } C = A \text{ and } (B \text{ and } C)$
 $(A \wedge B) \wedge C = A \wedge (B \wedge C)$

$(A \text{ or } B) \text{ or } C = A \text{ or } (B \text{ or } C)$
 $(A \vee B) \vee C = A \vee (B \vee C)$

would you like coke or orange juice
 $\begin{matrix} \uparrow & & \uparrow & & \uparrow \\ \text{operand} & & \text{operator} & & \text{operand} \\ \uparrow & & \uparrow & & \uparrow \\ A & & \vee & & B \end{matrix}$

I will not either take coke or orange juice. $= (A \vee B)'$

X = sky is clear

$\bar{X}$ = sky is not clear
 $\nwarrow$ operator

Implication $\% \quad$ 'X' is true and 'y' is false, then the result is false, otherwise it is true. It is represented by $\rightarrow / \Rightarrow$ symbol
or
if... then

X	Y	$X \rightarrow Y$
0	0	1
0	1	1
1	0	0
1	1	1

POS $\boxed{X \rightarrow Y = \bar{X} + Y}$

X = sky is clear $\} \quad X \rightarrow Y =$ if sky is not clear then it is raining.

$X = \text{sky is clear}$
 $Y = \text{It is raining}$

$X \rightarrow Y = \text{if sky is not clear then it is raining.}$
 $Y \rightarrow X = \text{if it is not raining then sky is clear.}$

$A = \text{'You practice for exam'}$

$B = \text{'You will succeed'}$

$A \rightarrow B = \text{'If you not practice for exam then you will succeed'}$

or
 $B \rightarrow A = \text{'If you will not succeed then you practice for exam'}$

if $\frac{5=7}{F}$ then $\frac{15=22}{F} \Rightarrow \text{True}$

if $x=2$ then $2x=4 \Rightarrow \text{True}$

if $\frac{x=2}{T}$ then $\frac{3x=5}{\text{Not possible}} \Rightarrow \text{False}$

if $x \neq 2$ then $\frac{3x=5}{\text{True}} \Rightarrow \text{True}$

if $\frac{5x > 15}{\text{false}}$ then $\frac{5x = 15}{\text{true}} \Rightarrow \text{True} \quad [\because x = 3]$

Equivalence : Both statements or variables have similar values to result into true; it is called "if and only if", iff.
 It is represented by ' $\leftrightarrow$ ' / ' $\Leftrightarrow$ ' symbol.

A	B	$A \leftrightarrow B$
0	0	1
0	1	0
1	0	0
1	1	1

$$\text{SOP} = A \leftrightarrow B = \bar{A} \cdot \bar{B} + AB$$

$$\text{POS} = A \leftrightarrow B = (A + \bar{B}) \cdot (\bar{A} + B)$$

$$A \leftrightarrow B = \bar{A} \bar{B} + AB$$

$A = \text{sky is clear}$

$B = \text{it is raining}$

$A \leftrightarrow B = \text{if sky is clear only if it is not raining}$
 operators

Contrapositive: $(A \rightarrow B) = (\text{not } B) \rightarrow (\text{not } A)$

$$A \rightarrow B = \bar{A} + B$$

A	B	A'	B'	$A \rightarrow B$	$\bar{B} \rightarrow \bar{A}$
0	0	1	1	1	1
0	1	1	0	1	1
1	0	0	1	0	0
1	1	0	0	1	1

$\uparrow = \uparrow$

$B' \rightarrow A'$
false $\rightarrow$ true = true

to get 'false' in implication, the rule is, first value must be 'true' and second value must be 'false'.

Tautology = When the resulting column have nothing but only '1'

A	B	logic
0	0	1
0	1	1
1	0	1
1	1	1

Contradiction = When the resulting column have nothing but only '0'

A	B	logic
0	0	0
0	1	0
1	0	0
1	1	0

Inverse: $P \rightarrow Q = \bar{P} \rightarrow \bar{Q}$
 $\nearrow$ antecedent
 $\nearrow$ consequent

Converse: $P \rightarrow Q = Q \rightarrow P$

Consistence Statement: is a logical statement which always generate 'true' or 'false' value as per given logic. For example $A \wedge B$, $A \vee B$.

De Morgan's Law:

$A \wedge B$, $A \vee B$..

De Morgan's law:

$$(i) (p \wedge q)' = \bar{p} \vee \bar{q}$$

$$(ii) (p \vee q)' = \bar{p} \wedge \bar{q}$$

Home work: Draw the truth table to prove De' Morgan's law.

'+' $\leftarrow$ conjunction

'.' $\leftarrow$ disjunction.

Note: In propositional logic, a contingency is a compound proposition, that is true in some cases and false in others, and therefore is neither tautology nor a contradiction.